

Dennis Chen

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EDUCATION

Cornell University

Master of Engineering in Computer Science

Bachelor of Science in Computer Science, GPA: 3.2

Ithaca, NY

Graduation Date: Dec. 2026

May 2026

WORK EXPERIENCE

Redtail Studios

AI Engineer Intern

Remote

Jul. 2026 – Present

- Designed an asynchronous, event-driven architecture using S3, SQS, Lambda, and EventBridge, enabling parallel processing across 18 platforms and ~2,000 records.
- Automated recurring market-data collection, eliminating up to 2.5 hours of manual scraping per historical refresh.
- Built 2 comprehensive test suites comprising 288 unit tests and 15 integration tests, simulating FastAPI calls and AWS services with Moto.
- Resolved an SQS production failure consuming 85% of monthly AWS capacity using timeouts, retries, S3 recovery, and dead-letter queues.
- Developed FastAPI endpoints with guest authentication and automated S3 data validation.

Cara AI / Altheros Capital

Full-Stack Developer Intern

Remote

Jun. 2026 – Present

- Built a Swift networking layer from scratch, centralizing API calls and automating authentication and refresh-token handling.
- Standardized REST API documentation across 33 endpoints using structured errors, versioning, and OpenAPI specifications.
- Implemented 46 automated tests across Swift networking, backend schemas, and database migrations; engineered GitHub Actions CI/CD checks to detect API specification drift.
- Identified 5 pre-release defects, including 2 data-loss risks, and implemented rollback protection before public release.
- Supported development of a landing page for another portfolio company, expanding full-stack development across the firm's product portfolio.

Gao Labs

Machine Learning Researcher

Ithaca, NY

Aug. 2025 – May 2026

- Reduced Integrated Squared Error by 75% by developing a ConvLSTM adaptive sampler with a diffusion-inspired training loop.
- Accelerated training throughput 5× on NVIDIA A100 clusters by optimizing I/O dimensions and increasing batch size from 8 to 40.
- Improved data efficiency by achieving comparable convergence with 23% fewer samples through progressive heterogeneity training.
- Led 7 engineers through weekly Agile sprints, converting research models into scalable training pipelines.

Applied Science and Technology Research Institute

Summer Intern

Hong Kong

Jun. 2025 – Aug. 2025

- Engineered a Python/OpenCV pipeline for ophthalmic scan reconstruction and visualization, reducing clinical analysis time by 50%.
- Developed Gaussian filtering and Canny edge-detection algorithms to isolate corneal structures across large-scale datasets.
- Automated image preprocessing and visualization to improve workflow consistency across 25 scans.
- Presented technical findings to senior leadership on automated computer vision for early-stage disease identification.

PROJECT EXPERIENCE

Discord Giveaway Bot | Python, PostgreSQL, Discord API, YouTube API, OpenAI API, Google Cloud

2026

- Developed a continuously running bot that searches YouTube hourly, processing up to 50 videos/query to identify legitimate giveaways.
- Reduced irrelevant processing by 80–90% by filtering approved creators before keyword and LLM classification.
- Architected a YouTube, PostgreSQL, and OpenAI workflow supporting up to 100 Discord servers.
- Automated distribution of 6–7 qualified videos/day, reaching 10–15 during game updates.
- Deployed on Google Compute Engine with persistent database-backed tracking.

Zulip Open-Source Contribution | Python, PostgreSQL, Django, Git

Feb. 2026 – May 2026

- Engineered Django/PostgreSQL backend logging infrastructure to capture granular engagement metrics for an AI summarization feature.
- Developed automated test coverage to validate logging integrity while maintaining zero impact on existing analytics modules.
- Managed GitHub pull requests through review, documentation, CI/CD troubleshooting, and iterative code changes in an Agile development environment.

Retrieval-Augmented Generation (RAG) System | C++, FAISS, llama.cpp

Nov. 2025 – Dec. 2025

- Architected a C++/FAISS retrieval pipeline supporting 10K+ document embeddings with llama.cpp inference.
- Optimized memory efficiency using Scalar Quantization and Vector Normalization for high-speed semantic retrieval.
- Integrated retrieval-based prompting to ground LLM responses and reduce hallucinations.

LEADERSHIP EXPERIENCE

Cornell University - Undergraduate Consultant (Fall 2023 – Fall 2025) → Undergraduate Teaching Assistant (Spring 2026) → Graduate Teaching Assistant (Fall 2026)

Ithaca, NY

- Evaluated 100+ Python scripts weekly, providing feedback on software development, debugging, and programming practices.
- Facilitated technical instruction for 180+ students through weekly recitations, translating complex computer science concepts into practical problem-solving frameworks.
- Supported programming assignments, examinations, and large-scale projects across 6+ semesters, maintaining consistent technical evaluation and instructional standards.

TECHNICAL SKILLS

Languages: Python, C++, Java, C, Swift, OCaml | Backend & APIs: FastAPI, Django, REST APIs, PostgreSQL, Alembic, OpenAPI

Cloud & Infrastructure: AWS (S3, SQS, Lambda, EventBridge), Google Cloud, Docker, NVIDIA A100, Linux

Software Engineering: Git, GitHub Actions, CI/CD, Automated Testing, Integration Testing, Distributed Systems, System Design

ML/AI: PyTorch, Scikit-learn, LLMs, RAG, LoRA, MCP, Diffusion Models, ConvLSTM, llama.cpp

Data & Computer Vision: FAISS, OpenCV, pandas, NumPy